# Text Summarization of News Reports

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## 1 Problem statement and motivation

In the modern era, where news can be presented in multiple forms, whether it is written by human or generated by machines. In this fast evolving world, there is a large amount of information and content being generated per second. Humans being Humans, Humans are lazy, we want any information or content we are consuming to be short and easy to digest.

Megari and Genova (2026) has noted that individuals exhibiting difficulties in reading and mathematics, often struggle with following instructions or retaining information. The authors have said that educators have noted children with these deficits in working memory often display inattentiveness and exhibit short attention spans. This similar behavior has been observing in some of my juniors, where they scoff at the idea of reading the newspaper, web articles or readable content in video games.

To help the future generation process the walls of text and learn, turning towards Natural Language Processing (NLP) was the right choice. By harnessing the power of NLP, we can automate this process by allowing machines to process and learn the human language and summarize the walls of text in a short amount of time.

With all of these factors in mind, we will be developing a NLP Pipeline to help summarize the long text contained in news articles. At the end of this, we will look at how this NLP Pipeline can be extended to be deployed in potential real life scenarios.

## 2 Research hypothesis

Using a Bidirectional and Auto-regression Transformers (BART) Model, which is an encoder decoder architecture, would be best for text summarization.

Similar architectural designs has proven to be superior in image segmentation tasks ((Minaee et al., 2021) ), which focus on extracting features and classifying the individual pixels in the image. By employing a similar architecture but modify it for the domain of language, by using an encoder only model like BERT (Bidirectional Encoder Representations for Transformers), to extract the features based on context from its surrounding ((Devlin et al., 2019)), and for the decoder part, using Generative Pre-trained Transformers (GPT), as it excels in generating human-like text ((Radford et al., 2018)). By combining the architectures of BERT and GPT, firstly we get BART, but the main point being that we can exploit the features of these architectures individually for the grand task of text summarization.

Along the way, we want to analyze and answer related research questions: 1. How big are news articles? 2. Why is it difficult to understand these articles? 3. Why are encoder decoder models popular? 4. How long does it take for the model to train, generate, and evaluate? 5. Where can we use the output of this model?

## 3 Related work and background

The main reason for choosing this topic was my intial observations of my juniors and their refusal and disinterest to read long form content. Hence, We embarked on a research journey to find out why. According to the authors of (Megari and Genova, 2026), millions of students are impacted by Attention-Deficit/Hyperactivity Disorder (ADHD) and Specific Learning Disorders (SLDs). They state that Executive Functions (EF), a range of

higher-level cognitive processes, develop until late adolescence but the important changes are during preschool years. This information combined with the fact of the rise of popularity of the internet and the consumption of content on there may have potentially led to this disinterest to read long form content.

New Outlets now a days tend to label their articles in a way that misaligned with readers' dominant motives, while the outlets themselves prioritize on improving their popularity using popular titles and tags (Dagogo-Jack and Watson, 2026). This misalignment will lead to students not bothering to even continue after being disinterested from the title. Hence, to help the students with being interested in new articles, we need to adjust the titles in a way that will make them interested, but also summarize it in a way that the reader will be engaged throughout the summarize article.

After going through the detailed survey regarding automated text summarization provided by (Mridha et al., 2021), which gave us a much deeper and insightful understanding of text summarization techniques, architectures, evaluation, and their advantages and disadvantages. But we will mainly be focusing on those sections where datasets are used related to news articles. There are 6 research papers that are of interest and are as follows:

1. The paper whose achieved a way to decrease summarization redundancy with DISCOBERT, where they used multiple policies such as BanditSum, NeuSum, JECS, BertSum, PNBert and HiBert. At the end, they were able to achieve their highest ROUGE 1,2 and L score as 50.00, 30.38 and 42.70 respectively. The authors of this paper want to explore better encoding methods in their future works (Xu et al., 2020).

2. The paper that did a quantitative and qualitative assessment of 15 algorithms such as word frequency, TF/IDF, Lexical Similarity, SL, SP, Text Rank etc. They achieved the following performance metrics with Recall, Precision and F measure as 0.47, 0.19, and 0.26 respectively (Ferreira et al., 2013).

3. The paper that implemented a joint extractive and compressive summarizer. This is a Neural network model which uses a sentence and document encoder, a compression encoder and a compression classifier. This model achieved ROUGE 1 and 2 scores of 45.5 and 25.3 respectively (Xu and Durrett, 2019).

4. The paper wanted to achieve text summarization by developing a semantic model, where they use a LSTM - CNN model. They have achieved a ROUGE 1 and 2 score as 34.9 and 17.8 respectively (Kapse and Deshmukh, 2025).

5. The paper that used an adversarial technique for abstractive text summarization which used a generative and discriminative model for human generated positive samples. They achieved ROUGE 1, 2, and L scores of 39.92, 17.65, and 36.71 respectively (Liu et al., 2018).

6. The paper that verified real consistency and identity conflicts using an abstractive neural model. Here they focused on swapping entities and pronouns, injecting noise and other such technique to ensure their model is capable of handling the same. Their model achieved an accuracy of 74.15 with an F1 score of 0.5106 (Narwadkar and Bagade, 2023).

With all the required background, information and knowledge gained, we can finally create a checklist of what all needs to be accomplished for our final task to be completed.

### 3.1 Accomplishments

- Task 1: Research Domain and finding problem - Completed
- Task 2: Finding well annotated and relevant datasets - Completed
- Task 3: Preprocess dataset - Completed
- Task 4: Build and Train Model (Baseline) - Completed
- Task 5: Build and Train Model (Actual) - Completed
- Task 6: Hyperparameter optimization and fine-tuning - Not Completed due to lack of computing resources
- Task 7: Model Evaluation (Computational) - Completed

- Task 8: Model Evaluation (Human Verification) - Not Completed due to lack of human resources
- Task 9: Error Analysis - Completed
- Task 10: Document results - Completed

## 4 Approach and Methodology

From previous experiences of using autoencoders in image processing and analysis, has shown and proven to me that they are superior in extracting features and classifying every part of the image. The encoder will extract the features of the image, while the decoders will reconstruct the image while classifying the Regions of Interest (ROIs) and pixels (as per configuration and training). If this knowledge base can be transferred to the domain of NLP, it would serve the same purpose where the encoders will be responsible for finding the semantic and syntactic context of the sentences and the decoder will be responsible for reconstructing the sentences into human readable form. We will modify the encoders to ensure that we get a summarized form rather than reconstructing the entirety of the original input.

For the Baseline, we shall be constructing a BART model, where the encoder uses BERT and the decoder uses GPT. Instead of training from scratch, we shall be using transfer learning to import pre-trained weights (we will use bart-base here as the pre trained weights), and train our existing dataset on top of this. The potential failure points of the model will be mainly due to the model not being trained for enough number of steps to produce a readable or meaningful sentence regardless of what our target summary it was originally trained on.

Regardless, we have successfully completed a working implementation of what we originally tasked out to do. The key components of our implementation are as follows:

1. We install all the required packages and libraries such as tarfile (extract the zipped dataset), os (for file operations), matplotlib, wordcloud (for plotting), torch (for constructing the neural network), evaluate, and NLTK (for evaluation purposes).

2. We setup universal variables such as the Random seed for torch and cuda (Here we use 42), hyperparameters such as batch size, epochs, and learning rate, and finally links and paths for the dataset and where the trained model will be saved. This is to ensure reproducibility of results and configurations as per device.

3. Using the tarfile library we extract the zip files and store it in a local instance and finally visualize and load the dataset.

4. Using BartTokenizer, we tokenize our dataset and prepare it for when it enters the training and model evaluation loop.

5. We load our BartForConditionalGeneration from the transformers group in torch, and load the pre trained weights from bart-base.

6. We initialize the optimizer, here it is AdamW (from past experience they work better with auto encoders), and set it to ensure that gradients from previous iterations are not taken forward.

7. In the actual training loop, we pass the input ids, attention mask, and labels into our model and calculate the loss. Using this loss, we will backpropagate and adjust the weights.

8. In the model evaluation loop, we generate the summarized text and compare them with the actual summarized text and calculate Recall-Oriented Understudy for Gisting Evaluation (ROUGE), Bilingual Evaluation Understudy (BLEU), and Metric for Evaluation of Translation with Explicit ORdering (METEOR), as these are commonly used metrics for lexical analysis (Mridha et al., 2021).

9. Finally we actually train the model and save it so it can be either retrained or deployed accordingly.

Parts of the current implementation where inspired by code from Week 7 of lab sessions (Sequence to Sequence models, GPT). Since the logic behind BART uses BERT as its feature extractor is important, the way the data is inputted into both of them must be similar.

Due to limitations of computing resources, We were able to run four instances of BART, one baseline (NLP_Bart-Baseline.ipynb) and three with different hyper-parameters (NLP_Bert-Summarization_Final.ipynb). Each run utilized approximately 65 GB worth of RAM running on an A100 GPU with high RAM mode, with each run

lasting between one to two hours. Along with testing and experimentation, this spent about 45 compute credits on Google Colab. Hence, the biggest challenge we faced were how to properly balance time and computational resources. There is nothing related to this that can be done for now, but for the next project onwards, we will ensure to calculate how much time and compute resources the model actually needs and schedule runs and experiments accordingly. Otherwise there were little to no issues when writing or running the code.

## 5 Dataset

This paper will be using the dataset used by "XL-SUM: Large-Scale Multilingual Abstractive Summarization for 44 Languages" published in Findings of the Association for Computational Linguistics (Hasan et al., 2021). This dataset is best suited for our task as it not only contains BBC news articles from different languages, but also contains the title, summary, and actual news article text, which is useful for the task we set forth to do (Our task will be to input the text field and get the summary field as output).

This dataset contains 45 different languages with the following straight forward JSON structure as demonstrated with an example: "id": "technology-17657859", "url": "https://www.bbc.com/news/technology-17657859", "title": "Yahoo files e-book advert system patent applications", "summary": "Yahoo has signalled it is investigating e-book adverts as a way to stimulate its earnings.", "text": "Yahoo's patents suggest users could weigh the type of ads against the sizes of discount before purchase. It says in two US patent applications that ads for digital book readers have been s̈less than optimal"o date. The filings suggest that users could be offered titles at a variety of prices depending on the ads' prominence They add that the products shown could be determined by the type of book being read, or even the contents of a specific chapter, phrase or word. The paperwork was published by the US Patent and Trademark Office late last week and relates to work carried out at the firm's headquarters in Sunnyvale, California. G̈reater levels of advertising, which may be more valuable to an advertiser and potentially more distracting to an e-book reader, may warrant higher discounts,ït states. Free books It suggests users could be offered ads as hyperlinks based within the book's text, in-laid text or even d̈ynamic contentsuch as video. Another idea suggests boxes at the bottom of a page could trail later chapters or quotes saying b̈rought to you by Company A.̈ It adds that the more willing the customer is to see the ads, the greater the potential discount. Ḧigher frequencies... may even be great enough to allow the e-book to be obtained for free,ït states. The authors write that the type of ad could influence the value of the discount, with l̈ower class advertising... such as teeth whitener advertisementsöffering a cheaper price than ḧighör m̈iddle classädverts, for things like pizza. The inventors also suggest that ads could be linked to the mood or emotional state the reader is in as a they progress through a title. For example, they say if characters fall in love or show affection during a chapter, then ads for flowers or entertainment could be triggered. The patents also suggest this could applied to children's books - giving the Tom Hanks animated film Polar Express as an example. It says a scene showing a waiter giving the protagonists hot drinks m̈ay be an excellent opportunity to show an advertisement for hot cocoa, or a branded chocolate bar.̈ Another example states: Ïf the setting includes young characters, a Coke advertisement could be provided, inviting the reader to enjoy a glass of Coke with his book, and providing a graphic of a cool glass.Ït adds that such targeting could be further enhanced by taking account of previous titles the owner has bought. 'Advertising-free zone' At present, several Amazon and Kobo e-book readers offer full-screen adverts when the device is switched off and show smaller ads on their menu screens, but the main text of the titles remains free of marketing. Yahoo does not currently provide ads to these devices, and a move into the area could boost its shrinking revenues. However, Philip Jones, deputy editor of the Bookseller magazine, said that the internet firm might struggle to get some of its ideas adopted. T̈his has been mooted before and was fairly well decried,ẗhe said. P̈erhaps in a limited context it could work if the merchandise was strongly related to the title and was kept away from the text. B̈ut readers - particularly parents - like the fact that reading is an advertising-free zone. Authors would also want something to say about ads interrupting their narrative flow."̈

For all languages, there is an 80% - 10% - 10% split for Train, Validation and Test Sets, except for in English where there is a 93% - 3.5% - 3.5% split

due to the shear volume of 329592 entries total. Furthermore, since the train and validation set for Scottish Gaelic, Kyrgyz and Sinhale had relatively low number of samples, their numbers were manually set to 500 samples for reliable evaluation.

For our paper, we shall be using the english dataset, but our solution can easily be extended to other languages by modifying the tokenization algorithm. The English dataset has 306522 samples for the training set, 11535 samples for the testing set, and 11535 samples for the validation set.

1, 2, and 3 contain the word clouds for the training, testing and validation datasets respectively. We can notice that the biggest words in all three of them are "said", "people", "will" and "say", this suggests that many of the articles contain conversation by others (i.e. the news article is quoting people), and the inclusion of people and will suggest promises made to the people (this pattern follows when going through some of the articles).

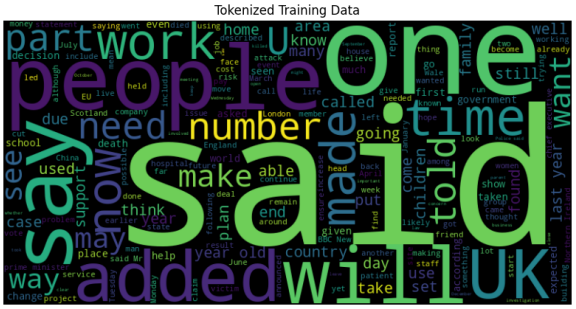



Figure 1: Word Cloud for Tokenized Training Set

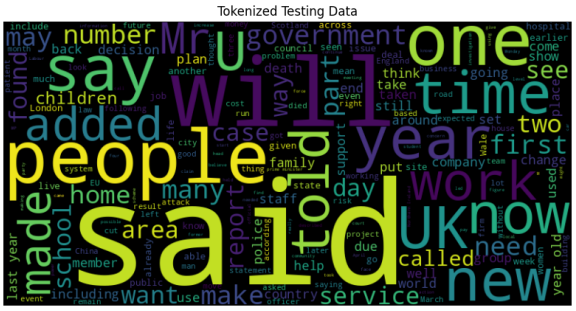



Figure 2: Word Cloud for Tokenized Testing Set

### 5.1 Dataset Preprocessing

To preprocessing the data so it can be used in a meaningful form by the BART Model, we shall be using a pre-trained BART Tokenizer from bart-base. The BART tokenizer (in the default setting) used is a sub-word tokenizer, which breaks words into smaller units to handle out of vocabulary words.

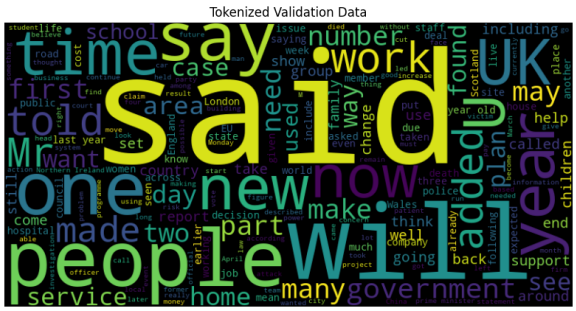



Figure 3: Word Cloud for Tokenized Validation Set

This is a crucial step, as summarization requires out of vocabulary words to form meaningful connections and the pre-trained tokenizer will help to achieve the same in a shorter period of time.

Other than this, there is no other requirement of preprocessing as the dataset in itself is clean and no further preprocessing will be needed.

## 6 Baselines

There were multiple baselines to consider from, but we chose the default BART Model from BartForConditionalGeneration with bart-base for the weights (using transfer learning). This was chosen as the baseline as with transfer learning, the model should provide good results. Furthermore, the weights used are trained for general purpose, making it the right choice for choose the same as the baseline.

## 7 Results, error analysis

### 7.1 Baseline Results

After setting our baseline and experimental models, we proceed to running the models. The results of the baseline models are as illustrated in 1. The ROUGE 1 score 20.27 suggests that 20.27% of individual words match, which is good as it is able to pick up some of the vocabalory. The METEOR score being 20.57% suggesting that it is able to pick up meaning but not the right wording structure as observed with the BLEU score of 0.80%. There is further proof that this model is unable to form coherent sentences due to the ROUGE 2 and 4 gram Precision score, as they are very low. Furthermore, the ROUGE L SUM (performance metric usually used for multi sentence summaries), with a score of 13.04 suggests that the model is poorly able to reconstruct the target summary.

A numerical analysis is one aspect, but we need to look at the actual text to check if it actually

| Sl. No. | Performance Metric | Result |
|---|---|---|
| 1 | ROUGE 1 | 20.27 |
| 2 | ROUGE 2 | 2.77 |
| 3 | ROUGE L | 13.05 |
| 4 | ROUGE L SUM | 13.04 |
| 5 | BLEU | 00.80 |
| 6 | Brevity Penalty | 1.0 |
| 7 | Precisions | [13.92, 1.51, 0.28, 0.06] |
| 8 | Length Ratio | 2.344 |
| 7 | METEOR | 20.56 |

Table 1: Performance Metrics for baseline implementation of BART

makes sense. The following is the sample output generated from the baseline model:

"By Kate DaileyBBC News Earlier this week, Trump posted a photo of himself sitting at a desk at Mar-a-Largo, a permanent marker hovering over a notepad. "Writing my inaugural address at the Winter White House, Mar-A-Lago, three weeks ago. Looking forward to"

As we can observe, the output summary generated is same as the input, this suggests that the baseline model is not doing any text summarization at all. And the numbers convey a similar message with respect to the target summary.

Now we move onto the other experiments.

### 7.2 Experiment 1

We will be using the following hyperparameters when training:

MAX_SUMMARY_LENGTH = 128
MAX_INPUT_LENGTH = 1024
BEAM_SEARCH_WIDTH = 4

After running these hyperparameters, we get the following results tabulated in 2. All the performance metrics are NULL suggesting that either the output generated from the model is empty or it is generating text that is utter nonsense with respect to the target summary.

The output generated from the Experiment 1 model is as follows:

""

We can observe that the output generated is indeed empty. This suggests that either the model is taking in too many padding values or empty strings beyond the input as the majority input, hence generating an empty string. This issue can be easily mitigated by reducing the MAX_INPUT_LENGTH to adjust it to the max length of the longest article, average length of the article or an arbitrary number.

| Sl. No. | Performance Metric | Result |
|---|---|---|
| 1 | ROUGE 1 | NA |
| 2 | ROUGE 2 | NA |
| 3 | ROUGE L | NA |
| 4 | ROUGE L SUM | NA |
| 5 | BLEU | NA |
| 6 | Brevity Penalty | NA |
| 7 | Precisions | NA |
| 8 | Length Ratio | NA |
| 7 | METEOR | NA |

Table 2: Performance Metrics for Experiment 1 of BART

### 7.3 Experiment 2 and 3

With the errors corrected from the previous experiment, we can proceed with the next two sets of experiments. The hyperparameters used for Experiment 2 and Experiment 3 are the same, with the only difference being the total number of samples being used are 10,000 and 306,522 respectively. The common Hyperparameters are as follows:

MAX_SUMMARY_LENGTH = 64
MAX_INPUT_LENGTH = 512
BEAM_SEARCH_WIDTH = 4

After running the model with these parameters, we get the following performance metrics tabulated in 3 and 4. The results showcased by Experiment 2 are similar to that of baseline indicating that the text produced is incoherent and not in any manner similar with respect to the target summary. But that is not a similar case with experiment 3, where we can see clear improvements compared to that of the other experiments (given, this model was trained for only one epoch). The ROUGE 1 and 2 score are 36.97% and 14.54% respectively, suggesting that there are more relevant number of single and double word pairs that match up with the target summary. The BLEU score is relatively low, but has improved significantly from baseline suggesting the model is learning to better match the sequence of words from the target summary. The Length Ratio has also reduced trying to adjust to get to 1.0, suggesting that the model is learning to generate the right length as the output. The 4 gram precision score (3.7%) has also improved suggesting that there are valid sequences of 4 words are matching. Finally the metric that matters most,

is that of ROUGE L and ROUGE L SUM, where the score has majorly improved indicating that the model is able to identify the features of the input text and summarize it correctly. But numbers can only convey soo much when talking about text.

| Sl. No. | Performance Metric | Result |
|---|---|---|
| 1 | ROUGE 1 | 20.28 |
| 2 | ROUGE 2 | 2.77 |
| 3 | ROUGE L | 13.058 |
| 4 | ROUGE L SUM | 13.057 |
| 5 | BLEU | 0.80 |
| 6 | Brevity Penalty | 1.0 |
| 7 | Precisions | [13.93, 1.515, 0.288, 0.069] |
| 8 | Length Ratio | 2.344 |
| 7 | METEOR | 20.575 |

Table 3: Performance Metrics for Experiment 2 of BART

| Sl. No. | Performance Metric | Result |
|---|---|---|
| 1 | ROUGE 1 | 36.97 |
| 2 | ROUGE 2 | 14.545 |
| 3 | ROUGE L | 29.275 |
| 4 | ROUGE L SUM | 29.279 |
| 5 | BLEU | 8.57 |
| 6 | Brevity Penalty | 0.77 |
| 7 | Precisions | [40.64, 14.019, 6.96, 3.7] |
| 8 | Length Ratio | 0.799 |
| 7 | METEOR | 29.37 |

Table 4: Performance Metrics for Experiment 3 of BART

The output generated by Experiment 2 is as follows:

"By Kate DaileyBBC News Earlier this week, Trump posted a photo of himself sitting at a desk at Mar-a-Largo, a permanent marker hovering over a notepad. "Writing my inaugural address at the Winter White House, Mar-A-Lago, three weeks ago. Looking forward to"

We can observe that the output is the same as the input, similar to that of the baseline. This suggests that the model, despite the number of samples used for training, it requires more number of samples for training to provide an actual impact for the output.

Now for the output generated by Experiment 3, it is as follows:

"US President-elect Donald Trump has delivered his inaugural address at Mar-a-Lago in Florida. "

From the output generated from experiment 3, we can clearly understand what the generated text is trying to convey, the model was able to extract the important features such as Mar-a-lago, and inaugural which are unique key features of the original news article. While there are a few anomalies like President-elect which does not make sense and is grammatically wrong. Looking at the other generated outputs, the summaries are providing a consistent summary with meaningful sentence, apart from a few words and symbols that appear out of nowhere where it does not fit. For a model that only ran for one epoch, this is a very good start for further training and fine tuning.

However, due to lack of computational resources, we are unable to proceed to fine tuning. But this current model is able to provide decent results with minimal training and finetuning.

## 8 Lessons learned and conclusions

We initially set out to create an NLP pipeline for summarization of news articles, which we achieved by training a BART model(Model Save File Name: Bart-Summarisation-Final-3.pth) with a ROUGE 1, 2, and L SUM of 36.97%, 14.545%, and 29.279% respectively. Compared to the original paper from which the dataset was obtained, where the same metrics were scored 37.601%, 15.1536%, and 29.8817% respectively. While we achieved similar results when training our model for only one epoch, we will get a better result if run for a few more epochs.

Furthermore, by completing this pipeline, it has shown that Autoencoder models can be used efficiently used in other domains by using appropriate techniques to transform the input into autoencoder calculable form. But do note that text summarization is a special task which is best aligned for autoencoders, hence we must make sure that the task shares the idea of the algorithm and architecture used.

Given ample amounts of computational and human resources, We would further like to keep running this model for more number of epochs and validate the output generated by humans or an expert. Furthermore, for finetuning, We would experiment

with modifying the current BART models and other transfer learning options. Furthermore, setting up an NLP pipeline to train from scratch will allow us to easily apply it to other languages and deploy the model as a browser plugin to live summarize text, so it is easier for students to grasp news articles and help develop a habit to read deeper if they are interested in the same.

In Conclusion, BART is a viable route for text summarization, with enough well annotated samples, the result will reflect the same.

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